Multidomain burden among midlife and older women across seven international ageing cohorts: a harmonized descriptive audit and cautionary profile-stability analysis

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Abstract

Background: Women's health research in later life increasingly needs multidomain measures that capture physical functioning, cognition, affective symptoms and chronic disease burden. Categorical profiles may help communicate heterogeneous burden patterns, but they can be overinterpreted when harmonization, complete-case selection, endpoint coupling and model stability are not explicit.
Methods: We analysed approved, downloaded, de-identified cohort datasets for women aged 50 years or older in CHARLS, ELSA, HRS, KLoSA, LASI, MHAS and SHARE. The authors cleaned the local cohort files, used harmonized ageing-study variables where available [1] and locally assembled domain inputs when required. Four within-cohort burden domains were oriented so that higher scores indicated worse burden. We treated Gaussian mixture models as descriptive partitions and audited measurement comparability, denominator attrition, LFO functional-change associations, algorithm concordance, formal secondary mortality models, all-sex interaction screens and prediction guardrails.
Results: The source screen included 79,938 women, of whom 76,293 had

complete four-domain descriptive assignments; the complete-domain 50–64-year stratum ranged from 45.4% to 67.8% across cohorts. Strict-core construction comprised CHARLS, ELSA, HRS and MHAS (29,058 women); SHARE, KLoSA and LASI remained sensitivity or descriptive tiers. All selected full-covariance four-domain Gaussian mixture solutions triggered near-singular covariance diagnostics, and cross-method agreement was limited in several cohorts. In strict-core LFO analyses, functional-change risk gradients were observed within cohort, but continuous three-domain scores matched or exceeded categorical LFO profiles for discrimination, model fit and decision-curve net benefit in 15 of 20 strict-core threshold comparisons. Secondary mortality Cox models showed high-burden class gradients, but proportional-hazards/time-drift flags in several cohorts and incomplete hard-endpoint harmonization kept mortality secondary. **Conclusions:** For midlife and older women's health research, the defensible contribution is a harmonized descriptive audit and cautionary profile-stability analysis using applied-for, downloaded cohort data cleaned by the authors. Categorical profiles may support burden mapping and communication, but the current evidence does not support stable latent endotypes, transportable categorical risk tools, population prevalence claims or prediction superiority over continuous domain scores. Continuous domain scores should remain the preferred analytic scale for functional-change modelling in this package.

Keywords: aging, women's health, multimorbidity, functional change, frailty, intrinsic capacity, harmonization

1 Background

Women's health research has historically emphasized reproductive health, cancer, mental health and selected chronic conditions, but health after midlife also depends on how functional ability, cognitive reserve, affective symptoms and chronic disease burden accumulate together. A women's health framing after midlife is therefore not limited to reproductive or cancer outcomes; it also includes capacity, disability, mental health and chronic-disease burden that shape independence and care needs. Functional ability, intrinsic capacity and multimorbidity are central to health-system planning for ageing populations [2–5]. Frailty phenotypes and deficit-accumulation indices summarize vulnerability [6–8], but they can compress clinically different domain patterns into a single severity continuum.

This study is intentionally scoped to women aged 50 years or older. This midlife-and-older threshold captures the transition into later-life health-risk accumulation while retaining comparability across the available ageing cohorts. Women have longer survival, higher late-life disability burden and different patterns of affective symptoms, multimorbidity and care needs [9]. The source cohorts did not provide a harmonized cross-cohort set of reproductive, menopausal or gynaecologic exposures for the present analysis; therefore the clinical question is not a sex-specific mechanism question. Instead, it asks how multidomain functioning, mental-health and chronic-disease burden should be described in women-focused ageing research when measurement equivalence and profile stability are uncertain. Exploratory all-sex LFO severity-by-sex interaction screens were fitted on a separately standardized all-sex scale; only one of six modelled cohorts had a nominal interaction signal, so these analyses do not support a consistent women-specific mechanism. The women-only design is therefore framed as a focused women's-health population study, not as evidence that the observed patterns are unique to women.

Recent multidomain trajectory work suggests that functional, cognitive, mood and chronic-disease indicators can evolve jointly [10]. The unresolved clinical and epidemiologic problem is whether categorical burden profiles remain interpretable after harmonization risk, complete-case selection, endpoint coupling and model instability are made explicit. We therefore focus on conservative description and claim calibration: whether multidomain burden can be summarized without overstating transportable latent subtypes or prediction performance.

2 Methods

2.1 Cohorts and analytic population

After registration or data-use application as required by each study, we downloaded de-identified cohort datasets from CHARLS, ELSA, HRS, KLoSA, LASI, MHAS and

SHARE official data access systems [11–17]. The authors cleaned the local cohort files, retained project-specific derived outputs and assembled the present analysis dataset. Variables included harmonized ageing-study products where available [1] and locally assembled domain inputs when no directly comparable cleaned variable was available. The analytic population was women aged 50 years or older at the cohort-specific selected baseline or analysis wave; we describe this population as midlife and older women rather than as a uniformly geriatric population. Sex coding was confirmed from local Working Data Stata value labels and do-files as `ragender=0` for women and `ragender=1` for men. Source-screen, complete-domain, descriptive-profile and functional-association denominators were separated in line with observational-study reporting principles [18]. Detailed cohort construction, harmonization and baseline covariate metadata are provided in Additional file 1.

2.2 Domain construction and harmonization review

Four domains were constructed: functional limitation, cognitive burden, affective symptoms and cardiometabolic/chronic disease burden. Functional items were anchored to activities of daily living and instrumental activities of daily living constructs where available [19, 20]. Affective symptoms used CES-D-family information where available [21]; SHARE used EURO-D information [22]. Scores were standardized within cohort and oriented so that higher values indicated worse burden. Retrospective harmonization review treated similar orientation as insufficient evidence for item identity or transportable measurement equivalence [23, 24].

2.3 Evidence tiers and descriptive profiles

The strict-core descriptive construction tier comprised CHARLS, ELSA, HRS and MHAS. SHARE was retained as construction-descriptive but validation-downgraded sensitivity evidence, KLoSA as functional-bridge sensitivity and LASI as baseline-only descriptive evidence. Cohort-specific Gaussian mixture models with two to five classes were fit to the four domain scores as descriptive clustering tools [25]. Models used full covariance matrices, `reg_covar=1e-6`, `max_iter=500`, `n_init=5` and random state 20260601. The selected model used the lowest BIC among converged models with minimum class size at least 5%, with class enumeration interpreted cautiously [26, 27]. The final stability guardrail used 200 nonparametric bootstrap refits per selected solution, covariance diagnostics and cross-algorithm agreement; these were treated as main guardrails rather than supplemental technicalities. Detailed profile dictionaries and stability outputs are provided in Additional file 2.

2.4 Functional-change association and unresolved endpoint guardrails

The functional-change endpoint was deterioration of at least 0.5 SD from baseline. Because functional change is a within-domain endpoint and the original four-domain descriptive profiles include baseline function, this analysis cannot be interpreted as independent hard-outcome validation. We therefore rebuilt exploratory LFO profiles after omitting the baseline functional domain and compared categorical LFO profiles with continuous cognitive, affective and cardiometabolic/chronic scores. Models adjusted for age, education, marital status, smoking and drinking. LFO model outputs are provided in Additional file 3. We additionally fitted formal secondary all-cause mortality Cox models, post-baseline hospitalization candidate models where cleaned binary headers existed, and calibration/decision-curve guardrails; these secondary endpoint and prediction guardrail outputs are provided in Additional file 4. Exploratory all-sex LFO severity-by-sex interaction models and survey-design audits are provided in Additional file 5. These analyses were treated as secondary or sensitivity evidence rather than harmonized primary validation.

3 Results

3.1 Denominator lock, baseline profile and complete-case selection

The source screen included 79,938 women aged 50 years or older, and 76,293 had complete four-domain descriptive assignments. The strict-core construction set comprised 29,058 women from CHARLS, ELSA, HRS and MHAS. The sensitivity/descriptive construction set comprised 47,235 women from SHARE, KLoSA and LASI. The complete-domain 50–64-year stratum ranged from 45.4% in HRS to 67.8% in CHARLS, supporting the midlife-and-older scope rather than a uniformly geriatric interpretation. These counts and percentages are analytic-sample summaries rather than survey-weighted population estimates. Complete-case selection was not neutral: excluded participants were older than included complete-domain participants in every cohort, including HRS (+10.4 years) and ELSA (+6.6 years). Table 1 and Figure 1 separate source, complete-domain, endpoint-availability and LFO model denominators. Each downstream analysis used its own endpoint- and covariate-complete denominator, so LFO, mortality and descriptive-construction denominators should not be interchanged.

Table 1 Baseline clinical profile and analytic denominators

Cohort	Tier/wave	Source/ complete N	LFO N/events/ follow-up	Baseline clinical profile	Excluded age delta
CHARLS	Strict-core; w1	6,878 / 6,019 (87.5%)	5,691; 1,766 (31.0%); 9.0 y	Age 61.9; 50–64/65+ 67.8/32.2%; BMI 23.9; chronic 0.7; 2+ chronic 17.6%; smoke/drink 6.5/15.0%	+4.2 y
ELSA	Strict-core; w1	6,292 / 6,104 (97.0%)	5,153; 1,316 (25.5%); 12.0 y	Age 65.1; 50–64/65+ 51.6/48.4%; BMI unavailable; chronic 0.7; 2+ chronic 14.9%; smoke/drink 18.2/85.2%	+6.6 y
HRS	Strict-core; w5	11,005 / 10,202 (92.7%)	9,431; 3,546 (37.6%); 15.0 y	Age 67.2; 50–64/65+ 45.4/54.6%; BMI 27.0; chronic 1.0; 2+ chronic 25.2%; smoke/drink 14.4/39.8%	+10.4 y
KLoSA	Bridge sensitivity; w3	4,344 / 4,081 (93.9%)	3,834; 1,144 (29.8%); 10.0 y	Age 66.0; 50–64/65+ 47.1/52.9%; BMI 23.4; chronic 0.7; 2+ chronic 17.4%; smoke/drink 3.0/17.7%	+4.0 y
LASI	Baseline-only; no wave field	28,165 / 27,433 (97.4%)	Unavailable in current cleaned pass	Age 62.6; 50–64/65+ 61.5/38.5%; BMI 23.1; chronic 0.6; 2+ chronic 13.5%; smoke/drink 3.7/2.6%	+6.5 y
MHAS	Strict-core; w1	7,440 / 6,733 (90.5%)	5,443; 1,437 (26.4%); 16.0 y	Age 62.4; 50–64/65+ 63.8/36.2%; BMI 27.7; chronic 0.7; 2+ chronic 13.0%; smoke/drink 9.5/17.6%	+4.5 y
SHARE	Validation down; w1	15,814 / 15,721 (99.4%)	12,205; 1,525 (12.5%); 11.0 y	Age 64.8; 50–64/65+ 53.9/46.1%; BMI 26.2; chronic 0.8; 2+ chronic 22.1%; smoke/drink 15.1/59.0%	+3.3 y

Source denominators are women aged 50 years or older in the selected baseline screen. Complete-domain denominators are four-domain descriptive assignments. The 50–64/65+ split and baseline clinical summaries are calculated among complete-domain participants unless unavailable. Follow-up years are nominal cohort intervals used for the functional-change endpoint. Endpoint availability is counted before the covariate-complete LFO modelling restrictions used in the LFO association analysis and Figure 1. Education, marital status, smoking, drinking and rural/region field availability is reported in Additional file 1.

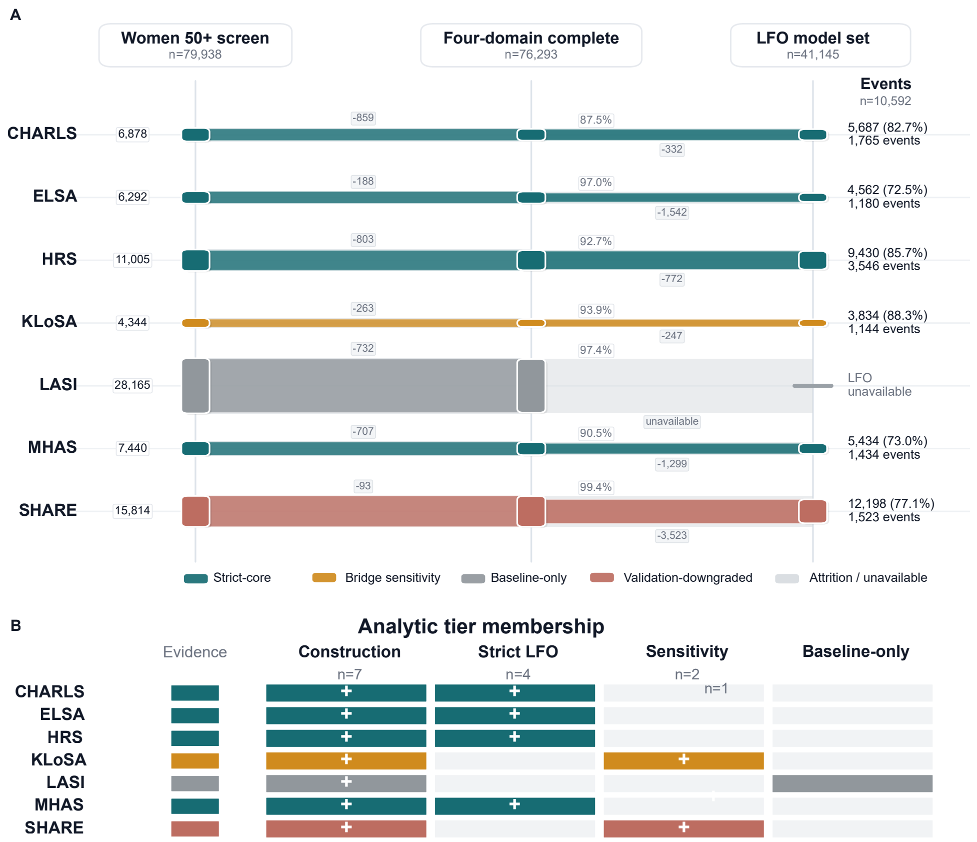

Fig. 1 Analytic denominator flow and evidence-tier membership. Panel A shows alluvial-style per-cohort attrition from women aged 50 years or older source screening to complete four-domain construction and covariate-complete LFO model denominators. These model-complete LFO denominators are intentionally smaller than the endpoint-available denominators in Table 1. Panel B summarizes analytic-tier membership for construction, strict-core LFO association, sensitivity and baseline-only descriptive roles. LASI remains baseline-only in the current cleaned-data pass.

3.2 Profile-stability guardrails

All selected full-covariance four-domain Gaussian mixture solutions triggered near-singular covariance diagnostics, with minimum covariance eigenvalues pinned at the regularization floor in at least one component per cohort. In the 200-refit bootstrap guardrail, all refits converged, but lower-tail agreement remained weak in several cohorts: p10 ARI was 0.35 in ELSA, 0.28 in HRS, 0.23 in KLoSA and 0.37 in SHARE.

Even cohorts with moderate-to-high median ARI were downgraded by near-singular covariance and weak lower-tail bootstrap agreement. The profile labels are therefore descriptive partitions for summarizing observed burden combinations, not stable latent disease entities or transportable risk tools (Table 2; Figure 2). The strict-core profile-pattern heatmap is retained as a supplementary descriptive display rather than as primary validation evidence.

Table 2 Model-stability guardrails for selected four-domain profile solutions

Cohort	Evidence tier	k	Bootstrap ARI	Cross-method ARI	Max condition	Claim status
CHARLS	Strict-core	3	1.00 (1.00; 0.43)	0.26 (0.12-1.00)	1.20e+06	Descriptive only
ELSA	Strict-core	5	0.84 (0.35; 0.23)	0.25 (0.22-1.00)	1.35e+06	Descriptive only
HRS	Strict-core	5	0.76 (0.28; 0.27)	0.20 (0.17-0.92)	1.52e+06	Descriptive only
KLoSA	Bridge sensitivity	3	0.98 (0.23; 0.21)	0.62 (0.22-0.93)	1.36e+06	Bridge sensitivity
LASI	Baseline-only	3	0.99 (0.97; 0.16)	0.22 (0.09-0.96)	1.19e+06	No follow-up validation
MHAS	Strict-core	5	0.98 (0.84; 0.33)	0.17 (0.15-0.95)	2.53e+06	Descriptive only
SHARE	Validation-downgraded	4	0.37 (0.37; 0.36)	0.22 (0.19-0.99)	9.28e+05	Downgraded

ARI = adjusted Rand index. Bootstrap ARI is median (p10; minimum) across 200 non-parametric refits. Cross-method ARI is median (range) against diagonal/tied GMM, k-means, sampled hierarchical clustering and continuous-severity tertiles. All selected full-covariance solutions triggered near-singular covariance diagnostics; therefore these profiles are descriptive partitions and not stable latent subtypes.

Fig. 2 Model-stability guardrails for selected descriptive profile partitions. Panel A summarizes 200-refit bootstrap ARI distributions, with higher values indicating greater refit stability. Panel B shows cross-method ARI against the selected full-covariance GMM across alternative clustering and continuous-severity comparators. Panel C ranks cohorts by log10 covariance condition number with the near-singular threshold shown as a reference line. Panel D maps bootstrap p10 ARI against log10 covariance condition to separate reproducible, fragile and near-singular descriptive solutions; it is a visualization of numerical guardrails, not a separate statistical model.

3.3 Exploratory LFO functional-change associations

Strict-core LFO functional-change analyses included 25,113 women and 7,925 functional deterioration events. Highest crude-risk LFO classes showed within-cohort functional-change gradients, although the ELSA highest-class contrast was not statistically conclusive. Continuous three-domain scores matched or outperformed categorical LFO profiles for discrimination and model fit in every strict-core cohort (Table 3;

Figure 3). Random-effects heterogeneity for the strict-core adjusted LFO risk ratios was considerable (pooled RR 1.40, 95% CI 1.19 to 1.64; I^2 80.2%; Q p=0.002; Additional file 3). With heterogeneity near 80%, the pooled estimate is a descriptive cross-cohort summary across non-identical instruments rather than a transportable effect estimate. These results describe within-cohort association with a functional change-score endpoint; they do not validate the four-domain descriptive profile patterns and do not demonstrate prediction superiority. SHARE and KLoSA are provided only as sensitivity rows because SHARE remained validation-downgraded and KLoSA used a functional bridge.

Table 3 Exploratory leave-functional-domain-out functional-change associations in strict-core cohorts

Cohort	LFO N/events	Risk contrast	Adjusted RR	Delta AUC	Reading
CHARLS	5,687/1,765 (31.0%)	C1 21.5% to C3 39.8%; RD 18.3 pp	1.63 (1.35 to 1.96)	-0.015	Continuous favoured
ELSA	4,562/1,180 (25.9%)	C1 21.1% to C5 42.2%; RD 21.1 pp	1.15 (0.96 to 1.37)	-0.007	Continuous favoured
HRS	9,430/3,546 (37.6%)	C1 28.4% to C4 51.8%; RD 23.4 pp	1.28 (1.16 to 1.41)	-0.008	Continuous favoured
MHAS	5,434/1,434 (26.4%)	C1 20.6% to C3 36.6%; RD 16.0 pp	1.61 (1.42 to 1.82)	-0.002	Continuous favoured

LFO = leave-functional-domain-out; RD = absolute risk difference between class 1 and the highest crude-risk LFO class. LFO classes were rebuilt without the baseline functional domain and are not the same objects as the four-domain descriptive profile patterns. Adjusted risk ratios use modified Poisson models with robust standard errors and adjustment for age, education, marital status, smoking and drinking. Delta AUC is profile AUC minus continuous-score AUC; negative values favour continuous three-domain scores. Full profile and continuous AUC values, SHARE/KLoSA sensitivity rows and class-level risks are provided in Additional file 3.

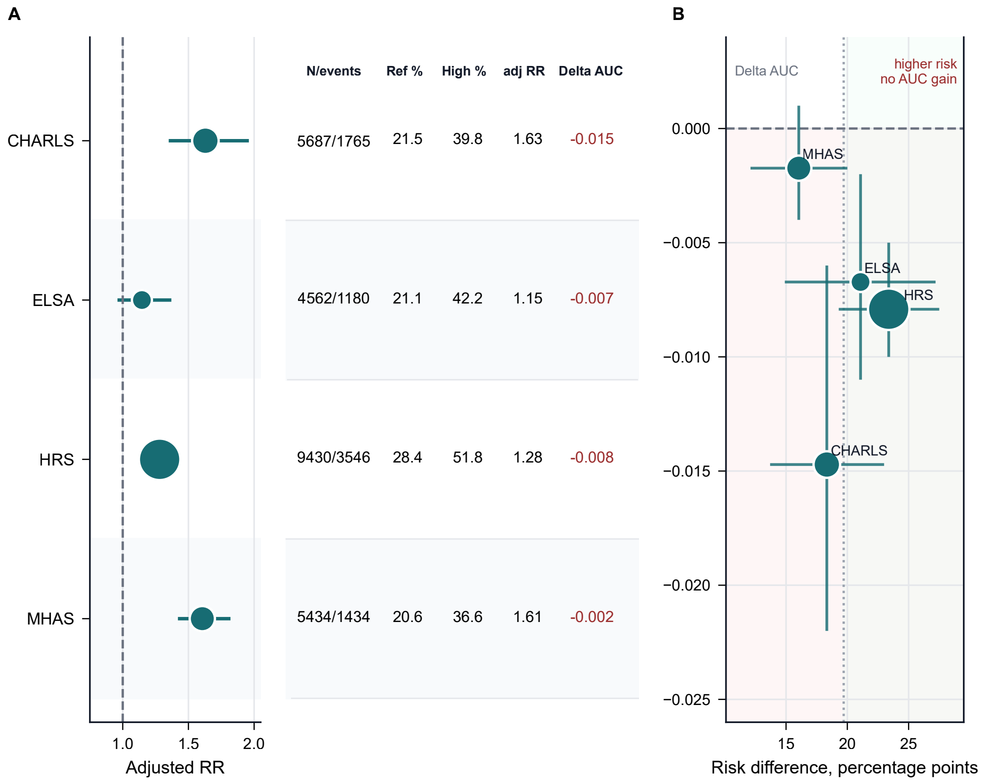

Fig. 3 Exploratory strict-core LFO functional-change associations. Panel A shows adjusted risk ratios comparing the highest crude-risk LFO class with the reference class, with point size scaled by model denominator and the adjacent table reporting N/events, reference-class risk, highest-class risk, adjusted RR and delta AUC. Panel B places cohorts in risk-difference versus delta-AUC space; horizontal intervals show uncertainty in crude risk difference and vertical intervals show 95% bootstrap intervals for delta AUC. Delta AUC is profile AUC minus continuous-score AUC, so negative values favour continuous three-domain scores.

3.4 Secondary mortality and prediction-guardrail analyses

Formal secondary all-cause mortality Cox models were fitted in six cohorts; LASI remained unavailable for mortality follow-up in the current cleaned pass. Highest-burden profile classes were associated with higher all-cause mortality in the modelled cohorts, but proportional-hazards or time-drift guardrail flags appeared in ELSA, HRS, KLoSA and SHARE. The ELSA death count was low relative to the nominal follow-up interval and may reflect incomplete mortality ascertainment in the current cleaned pass, so the ELSA hazard ratio should be read with particular caution.

Among strict-core cohorts, random-effects heterogeneity for the highest-class mortality HR was considerable (pooled HR 2.37, 95% CI 1.99 to 2.82; I^2 78.7%; Q p=0.003; Additional file 4). With considerable heterogeneity and PH/time-drift flags in several cohorts, the pooled HR and flagged cohort-specific HRs are descriptive guardrail summaries rather than constant-hazard or transportable effect estimates. Mortality is therefore reported as a secondary non-circular endpoint guardrail rather than as primary validation evidence (Table 4). Decision-curve guardrails also supported the main cautionary interpretation: continuous three-domain scores had higher apparent in-sample net benefit than categorical LFO profiles in 15 of 20 strict-core threshold comparisons, without optimism correction. Calibration outputs are likewise reported only as apparent in-sample diagnostics and are not interpreted as validated calibration.

3.5 Harmonization, hard-outcome and survey-design guardrails

The harmonization matrix remained central to interpretation (Figure 4). Functional-domain strictness differed across cohorts: CHARLS used IADL-only information, HRS used ADL-only information, KLoSA used a bridge proxy and SHARE remained validation-downgraded. Cognitive batteries were cohort-specific, SHARE affective symptoms used EURO-D, and cardiometabolic/chronic disease indicators differed in lipid/cholesterol availability. Raw/codebook/dofile auditing identified candidate weight, primary sampling unit and strata triplets in HRS, KLoSA, LASI and SHARE, but no codebook-confirmed harmonized seven-cohort survey-design triplet was available, so no survey-weighted prevalence claim is made. The all-sex interaction review found one nominal LFO severity-by-sex interaction among six modelled cohorts, which supports the focused women-only scope but not a consistent women-specific mechanism. Hospitalization candidate models were feasible only where cleaned binary headers existed and remain header-only sensitivity evidence until codebook mapping is completed; institutionalization and care-dependence endpoints remained unavailable.

Table 4 Secondary all-cause mortality Cox guardrail models

Cohort	Tier	N/deaths; time	Highest class HR	PH/time flag	Reading
CHARLS	strict-core	5,868; 703 deaths (12.0%); 9.0 y	C3 HR 1.82 (1.54 to 2.16)	No flag	Secondary
ELSA	strict-core	4,639; 317 deaths (6.8%); 12.0 y	C5 HR 3.07 (2.15 to 4.39)	Flagged	Time-averaged
HRS	strict-core	10,043; 5,569 deaths (55.5%); 15.9 y	C5 HR 2.60 (2.34 to 2.89)	Flagged	Time-averaged
KLoSA	bridge sens.	3,990; 726 deaths (18.2%); 10.0 y	C3 HR 1.30 (1.09 to 1.56)	Flagged	Bridge sensitivity
LASI	baseline-only	unavailable	NA	NA	Not modelled
MHAS	strict-core	6,477; 2,231 deaths (34.4%); 17.0 y	C3 HR 2.39 (2.10 to 2.72)	No flag	Secondary
SHARE	validation-down	12,952; 3,198 deaths (24.7%); 11.0 y	C4 HR 2.70 (2.40 to 3.03)	Flagged	Time-averaged

These are formal secondary all-cause mortality Cox models using available death timing fields after local data cleaning. They are reported as non-circular endpoint guardrails, not as primary validation, because proportional-hazards or time-drift flags were present in several cohorts, survey-design variables were not applied, and LASI mortality follow-up was unavailable in the current cleaned pass. The ELSA death count appeared low relative to nominal follow-up and may reflect incomplete mortality ascertainment in the current cleaned pass; the ELSA HR should therefore be interpreted with particular caution. HRs in flagged cohorts should be read as conventional time-averaged guardrail summaries rather than constant-hazard effects. Technical flag coding and AIC guardrail details are provided in Additional file 4.

These guardrails inform the clinical interpretation and claim boundaries summarized in Box 1 in the Discussion. The full workflow and supplementary profile displays are embedded after the references as Supplementary Figures S1-S5.

4 Discussion

This manuscript is deliberately scoped as a harmonized descriptive audit and cautionary profile-stability analysis for midlife and older women's health research. The seven-cohort comparison shows that multidomain burden among women aged 50 years or older can be summarized and audited, but the selected categorical partitions are

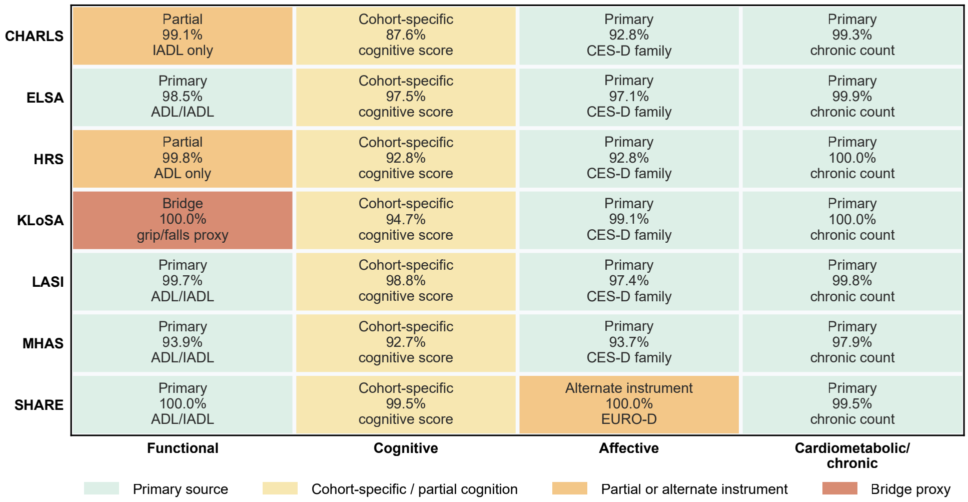

Fig. 4 Cohort-domain harmonization risk matrix. Cells summarize source tier, non-missing percentage and the principal construct used for each burden domain. The matrix is measurement-risk evidence rather than a biological result.

not robust enough to be interpreted as stable endotypes or transportable risk tools. The main positive contribution is transparent measurement and denominator mapping across several major ageing cohorts; the main cautionary finding is that categorical profile solutions are numerically fragile and do not improve on continuous domain scores for the current functional-change endpoint.

Box 1. Clinical interpretation and claim boundaries for women's health research

Evidence anchor	Interpretation	Boundary
Table 1 and Figure 1	A large midlife-and-older women analytic sample can be described with explicit denominator attrition and tier membership.	Counts are analytic-sample summaries, not survey-weighted population prevalence estimates.
Table 2 and Figure 2	Categorical profiles can summarize multidomain burden patterns and support communication.	Full-covariance GMM solutions were numerically fragile and should not be treated as stable latent endotypes or transportable risk tools.
Table 3, Figure 3 and Additional file 3	Continuous LFO cognitive, affective and cardiometabolic/chronic scores matched or exceeded categorical profiles for functional-change modelling.	The analysis does not show categorical prediction superiority; continuous domain scores remain the preferred analytic scale in this package.
Table 4, Figure 4 and Additional files 4-5	Secondary mortality gradients and harmonization/survey-design audits provide clinically relevant guardrails for further work.	Mortality remains secondary, hard-endpoint harmonization is incomplete and the all-sex interaction screen does not establish a consistent women-specific mechanism.

LFO = leave-functional-domain-out. This box is a discussion-level claim-calibration summary, not a separate statistical model.

The functional-change analysis should be interpreted conservatively. The LFO approach reduces predictor-side endpoint coupling by removing baseline function from the profile-construction step, but the outcome remains a within-domain change score rather than an independent hard clinical endpoint. The observed risk gradients therefore support exploratory within-cohort association, not validation of the four-domain descriptive patterns. This limitation is especially important because the formal secondary mortality analyses show that clinically more independent endpoints are partly feasible, while also demonstrating why mortality must remain secondary in this package because several cohorts triggered proportional-hazards or time-drift guardrails.

The women-only scope is also conservative. The analysis addresses midlife and older women as a clinically important population with high late-life disability burden, but it does not establish sex-specific mechanisms. Exploratory all-sex LFO severity-by-sex interaction models were fitted on a separately standardized all-sex scale and showed no consistent cross-cohort interaction signal; only CHARLS had a nominal interaction. These models were not used for women-only profile construction or outcome modelling. A stronger mechanistic women's-health paper would require a prespecified all-sex design, formal sex-interaction estimands and harmonized women-specific exposures. The present implication is narrower: women's health studies after midlife should report multidomain burden with transparent denominator, harmonization and stability guardrails, and should retain continuous domain scores when modelling functional change.

5 Strengths and limitations

Strengths include the seven-cohort scope, codebook-confirmed sex coding, explicit denominator locking, separation of strict-core primary evidence from sensitivity tiers, baseline clinical characterization, included-versus-excluded missingness auditing, item-level harmonization review, LFO functional-change modelling, formal secondary mortality modelling, all-sex interaction screens and 200-refit algorithm-stability diagnostics. Limitations remain substantial. Domain measures were harmonized by orientation and within-cohort standardization but were not instrument-identical across cohorts; this is particularly important for Table 3 because functional instruments differed by cohort. Complete-case selection likely removed older and higher-burden participants. Survey weights, strata and primary sampling units were screened using cleaned headers, Stata metadata and dofile/codebook text, but no common codebook-confirmed seven-cohort design triplet was available. LASI lacked a follow-up validation denominator, SHARE was validation-downgraded and KLoSA used a functional bridge. Mortality was modelled only as secondary evidence; hospitalization models remain header-only candidate analyses; institutionalization and care-dependence endpoints remain unavailable. Box 1 consolidates the resulting claim boundaries.

6 Conclusions

Across seven international ageing cohorts, multidomain burden among midlife and older women can be described, tiered and audited. The conclusion is cautionary: categorical profiles are descriptive partitions with important measurement, stability and endpoint-coupling caveats. For functional-change modelling in this package, continuous domain scores remain the preferred analytic scale.

Abbreviations

ADL, activities of daily living; AIC, Akaike information criterion; ARI, adjusted Rand index; BIC, Bayesian information criterion; CES-D, Center for Epidemiologic Studies Depression Scale; CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; GMM, Gaussian mixture model; HRS, Health and Retirement Study; IADL, instrumental activities of daily living; KLoSA, Korean Longitudinal Study of Aging; LASI, Longitudinal Aging Study in India; LFO, leave-functional-domain-out; MHAS, Mexican Health and Aging Study; SHARE, Survey of Health, Ageing and Retirement in Europe.

Declarations

Ethics approval and consent to participate

This study used approved, downloaded, de-identified cohort datasets from CHARLS, ELSA, HRS, KLoSA, LASI, MHAS and SHARE after registration or data-use application as required by each cohort. Each source cohort obtained ethics approval and participant consent according to its protocol. The authors cleaned and harmonized the analysis files locally. The present analysis did not involve direct participant contact.

Consent for publication

Not applicable.

Availability of data and materials

The source cohort datasets are available through their respective official data portals or data access systems subject to registration, application approval and data-use conditions. The authors downloaded, cleaned and harmonized the approved de-identified cohort files for this analysis. Source and derived participant-level data cannot be redistributed if doing so would violate source cohort data-use agreements. Analysis code, harmonization metadata and aggregate non-identifying outputs used to reproduce the manuscript tables and figures are archived at Zenodo: DOI 10.5281/zenodo.20589522, with the development version available at GitHub: https://github.com/Luciky-Leo/older-women-multidomain-burden.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

FL, JC and JS contributed equally to this work and share first authorship. FL contributed to study design, data cleaning, harmonization, statistical analysis, table and figure assembly and drafting of the manuscript. JC contributed as a co-first author to study design discussion, clinical interpretation of functional-change and secondary endpoint findings, assessment of clinical relevance, manuscript drafting and critical revision. JS contributed as a co-first author to the women's-health framing, obstetrics-and-gynaecology clinical interpretation, literature interpretation, manuscript drafting and critical revision. RG conceived and supervised the study, led the statistical strategy with FL, verified the analyses, finalized the manuscript and serves as the primary corresponding author. LL contributed clinical supervision, interpretation of women's-health implications and critical revision of the manuscript and serves as a co-corresponding author. All authors read and approved the final manuscript.

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Use of artificial intelligence tools

Large language model tools were used for drafting support, code-execution log summarization and language editing. No artificial intelligence tool was listed as an author. The authors verified all analyses, references, figures and manuscript content and take full responsibility for the final work.

Additional files

Additional file 1. File name: `additional_file_1_harmonization_and_cohort_construction.xlsx`. File format: XLSX. Title: Cohort construction, harmonization and baseline covariate metadata. Description: Workbook containing item-level harmonization crosswalks, cohort tier lock, harmonization risk matrix, midlife/older age distribution and baseline clinical/design/covariate availability tables.

Additional file 2. File name: `additional_file_2_profile_stability_and_descriptive_profiles.xlsx`. File format: XLSX. Title: Descriptive profile and model-stability guardrails. Description: Workbook containing GMM stability summaries, 200-refit bootstrap robustness tables, covariance diagnostics, selected class dictionary, descriptive profile summaries, profile-stability guardrail tables, algorithm robustness sensitivity and sensitivity-cohort profile support.

Additional file 3. File name: `additional_file_3_lfo_functional_change_associations.xlsx`. File format: XLSX. Title: Leave-functional-domain-out functional-change association outputs. Description: Workbook containing decoupled validation comparisons, functional endpoint leakage audit, strict-core LFO association table, sensitivity rows, class-level risks, AUC bootstrap intervals and cross-cohort heterogeneity summaries for functional-change analyses.

Additional file 4. File name: `additional_file_4_secondary_endpoints_and_prediction_guardrails.xlsx`. File format: XLSX. Title: Secondary endpoint and prediction guardrail outputs. Description: Workbook containing formal mortality guardrail models, hospitalization candidate models/status tables, hard-endpoint feasibility matrix, calibration metrics, decision-curve summaries, secondary endpoint prediction guardrails and cross-cohort mortality heterogeneity summaries.

Additional file 5. File name: `additional_file_5_survey_design_and_sex_comparator_audits.xlsx`. File format: XLSX. Title: Survey-design and sex-comparator audits. Description: Workbook containing survey-design variable and raw/codebook audits, survey triplet status, baseline male-comparator summaries/contrasts, all-sex LFO sex-interaction terms and a reviewer-risk summary sheet.

Supplementary Figures S1-S5 are embedded after the references in this manuscript file and are not separate tabular Additional files.

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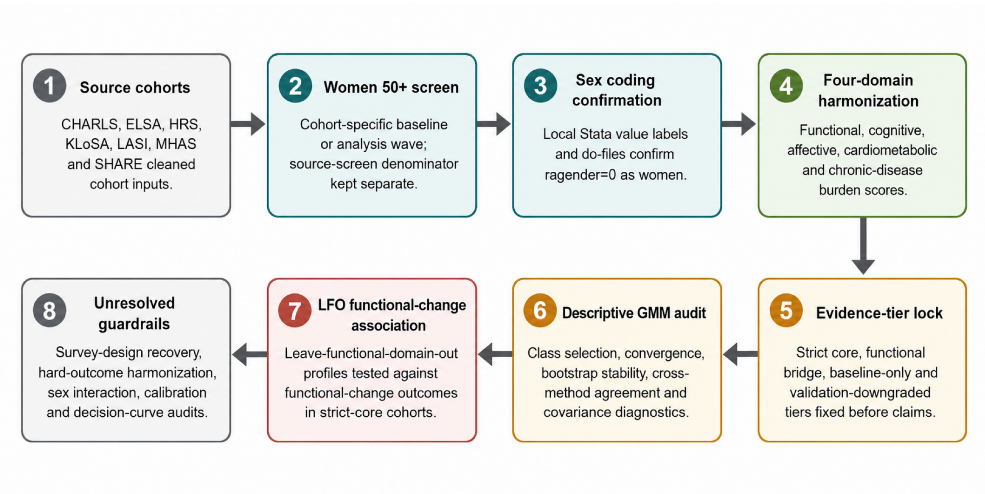
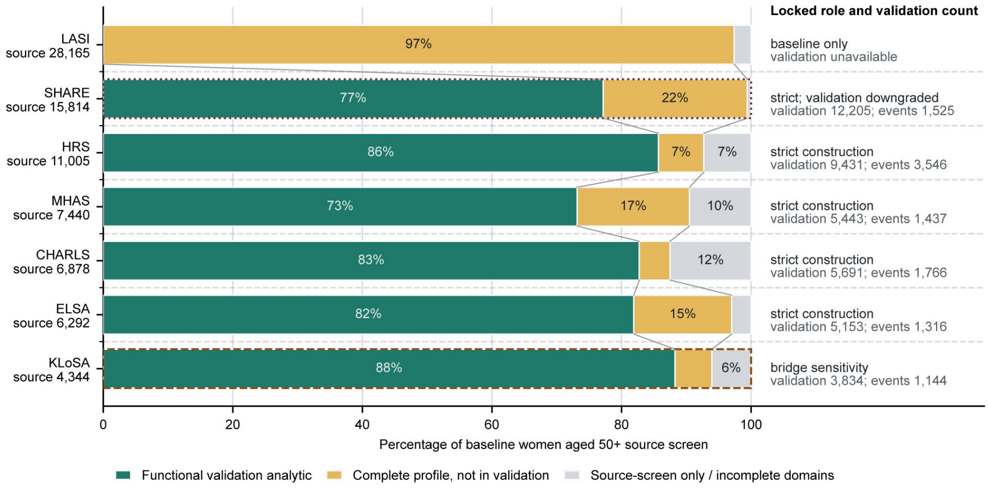
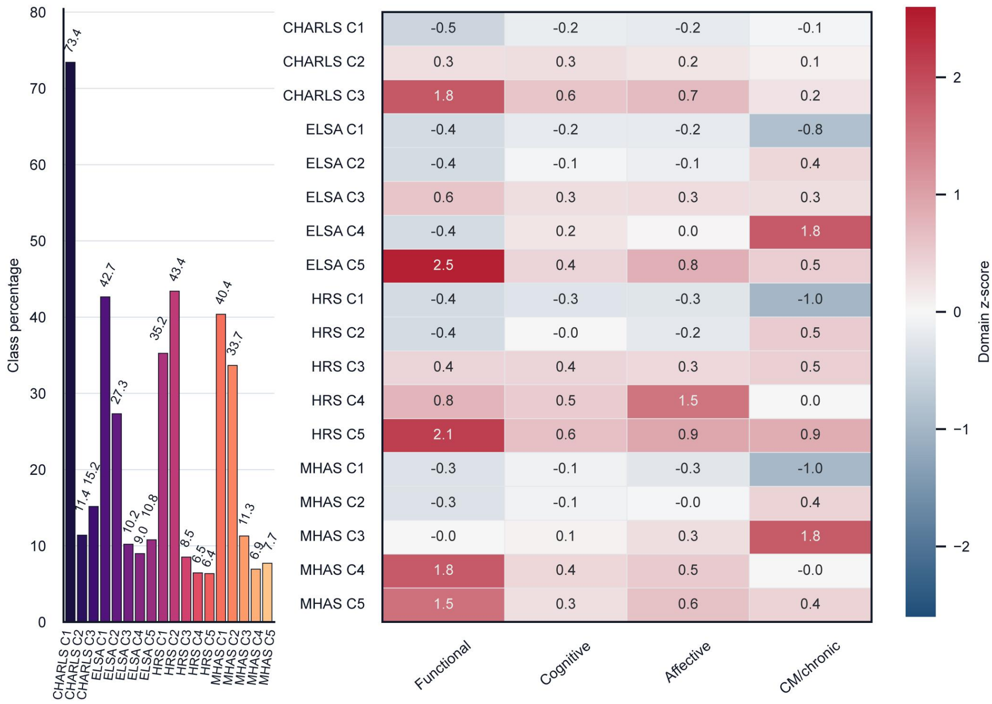
Supplementary figures



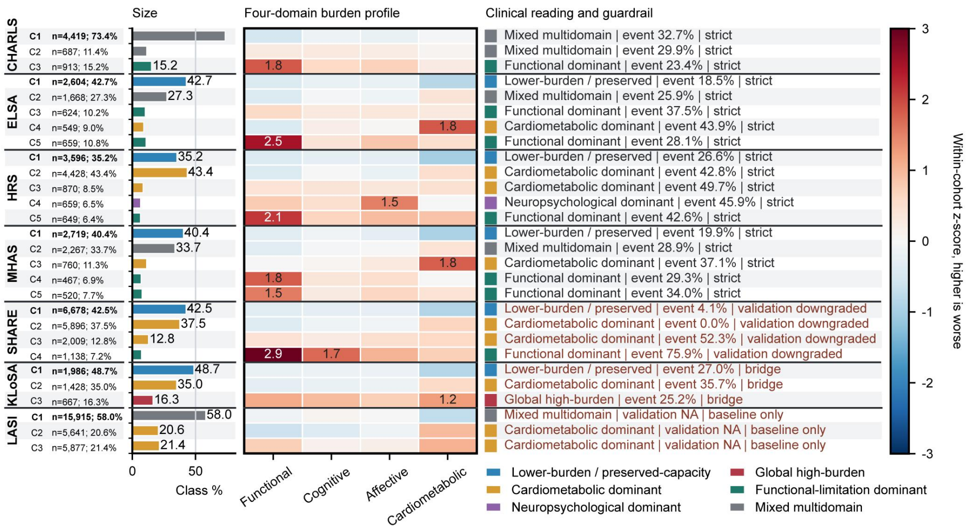
Supplementary Figure S1. Workflow schematic showing the analysis chain from source cohorts through women aged 50 years or older screening, sex-coding confirmation, four-domain harmonization, evidence-tier locking, descriptive GMM audit, leave-functional-domain-out functional-change association and unresolved guardrail audits.



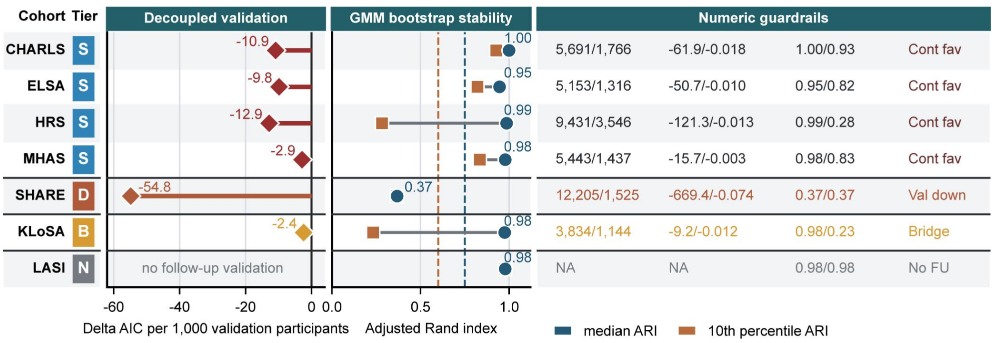
Supplementary Figure S2. Cohort denominator and validation dashboard showing source women aged 50 years or older, complete-domain analytic proportions, LFO validation availability and evidence-tier labels.



Supplementary Figure S3. Strict-core descriptive profile heatmap and class-size display for CHARLS, ELSA, HRS and MHAS.



Supplementary Figure S4. Full descriptive profile heatmap and clinical-reading panel across strict-core, sensitivity and baseline-only tiers.



Supplementary Figure S5. Decoupled validation, GMM bootstrap stability and numerical guardrail dashboard retained as supplementary context.